Coherent Spectral Codes: Passive Decoherence Resistance from Self-Aligning Topologies

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Abstract

We extend spectral codes—quantum error-detecting codes based on graph Laplacian eigenspaces—by incorporating insights from self-aligning topology theory. While Part I established that single-vertex errors are detectable via eigenvalue measurement, we now show that *eigenspace stability* provides a complementary form of *passive* protection against small perturbations. The key result is that random regular graphs, which exhibit maximal eigenspace stability under perturbation, yield spectral codes with natural decoherence resistance. We prove that the Davis-Kahan bound translates eigenspace stability into bounds on state fidelity under noise. Combined with active error detection, this gives a dual protection scheme: (1) large errors that change the eigenvalue are detected, while (2) small errors that preserve the eigenvalue cause minimal state disturbance due to eigenspace stability. We call codes satisfying both properties *coherent spectral codes*. Numerical experiments demonstrate that random 4-regular graphs achieve 90% eigenspace stability versus 33% for complete graphs, predicting significantly longer coherence times. This work bridges spectral graph theory, quantum error correction, and the mathematical theory of self-reference.

Keywords: quantum coherence, decoherence resistance, spectral codes, self-aligning topologies, random regular graphs, eigenspace stability

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1 Introduction

1.1 The Coherence Problem

Quantum coherence—the ability of quantum states to maintain superposition—is fragile. Environmental interactions cause *decoherence*, collapsing quantum information into classical mixtures. The standard approach to protecting quantum information uses *active* error correction: repeatedly measure error syndromes and apply corrective operations.

But active error correction has overhead: it requires ancilla qubits, measurement, and classical processing. An alternative is *passive* protection: encoding information in states that are inherently resistant to noise. Topological quantum computing pursues this via non-local encoding in anyonic states [3].

This paper develops a different approach to passive protection based on *eigenspace stability*. We show that encoding quantum information in the eigenspaces of graph Laplacians on *self-aligning topologies* provides natural decoherence resistance—small perturbations minimally disturb the encoded state.

1.2 Prior Work

In Part I [1], we introduced *spectral codes*: given a graph G with Laplacian L and eigenvalue λ, the code space is $E_\lambda = \ker(L - \lambda I)$. We proved:

1. Single-vertex errors map states between eigenspaces (Theorem 4.2)
2. Error detection via eigenvalue measurement (Theorem 4.3)
3. Distance bound $d \geq \lceil \mathrm{girth}(G)/2 \rceil$ (Theorem 5.1)

In separate work [2], we established that Laplacian eigenvectors are *fixed points* of self-modeling operators $M = g(L)$, and identified *self-aligning topologies*—graphs whose eigenvectors are stable under local perturbation. The key finding: **random regular graphs outperform highly symmetric structures** in eigenspace stability.

This paper synthesizes these results to address the coherence problem.

1.3 Main Contributions

1. **Dual protection framework** (Section 3): We identify two complementary protection mechanisms—active (error detection) and passive (eigenspace stability)—and show how they combine.

2. **Stability-fidelity theorem** (Section 4): We prove that eigenspace stability bounds state fidelity under perturbation, quantifying passive protection.

3. **Optimal topology selection** (Section 5): We show that random regular graphs provide optimal passive protection, outperforming highly symmetric alternatives.

4. **Coherent spectral codes** (Section 6): We define codes satisfying both active and passive protection criteria, and analyze their performance.

2 Background

2.1 Spectral Codes (Review)

Definition 2.1 (Spectral Code [1]). For graph G with Laplacian L and eigenvalue λ, the spectral code is:

$$\mathcal{C}(G, \lambda) = E_\lambda = \ker(L - \lambda I) \tag{1}$$

Theorem 2.2 (Error Detection, Part I). *For interior eigenvalue λ (i.e., $0 < \lambda < \lambda_{\max}$), every single-vertex error $Z_i = I - 2|i\rangle\langle i|$ is detected: measuring L after Z_i yields eigenvalue $\mu \neq \lambda$ with positive probability.*

This provides *active* protection: errors are detected and can be corrected.

2.2 Self-Aligning Topologies (Review)

Definition 2.3 (Self-Alignment Score [2]). For graph G, the self-alignment score measures eigenspace stability under local perturbation:

$$S(G) = \alpha \cdot \frac{\lambda_1}{\lambda_{\max}} + \beta \cdot \mathbb{E}[|\langle v_i, v_i' \rangle|] + \gamma \cdot (1 - \sigma_{\text{stability}}) \tag{2}$$

where the expectation is over random local perturbations and eigenvector indices.

Theorem 2.4 (Davis-Kahan [4]). *Let L and $L' = L + E$ be Laplacians with eigenvectors v and v' for eigenvalue λ. If $\delta = \min_{\mu \neq \lambda} |\lambda - \mu|$ is the spectral gap to adjacent eigenvalues, then:*

$$\sin \angle(v, v') \leq \frac{\|E\|_2}{\delta} \tag{3}$$

Proposition 2.5 (Random Regular Graphs Excel [2]). *Random d-regular graphs achieve highest eigenspace stability (0.90) among tested topologies, compared to complete graphs (0.33), hypercubes (0.55), and cycles (0.65).*

The key insight: **structured randomness outperforms pure symmetry** for eigenspace stability.

3 Dual Protection Framework

We now formalize how active and passive protection complement each other.

3.1 Error Classification

Consider a noise channel $\mathcal{N}$ acting on code state $|\psi\rangle \in E_\lambda$. Decompose the noise into:

Definition 3.1 (Error Components)**.** For noise operator N, define:

$$N_\parallel = P_\lambda N P_\lambda \quad \text{(within-eigenspace component)} \tag{4}$$

$$N_\perp = (I - P_\lambda) N P_\lambda \quad \text{(eigenspace-changing component)} \tag{5}$$

where P_λ is the projector onto E_λ.

The total effect is $N|\psi\rangle = N_\parallel|\psi\rangle + N_\perp|\psi\rangle$.

3.2 Two Protection Mechanisms

1. **Active protection (detection):** The component $N_\perp$ moves the state outside E_λ. Measuring L detects this: eigenvalue $\mu \neq \lambda$ indicates error. This is Theorem 2.2.

2. **Passive protection (stability):** The component $N_\parallel$ keeps the state in E_λ but may distort it within the eigenspace. Eigenspace stability bounds this distortion.

Proposition 3.2 (Complementary Protection)**.** *For noise $N = N_\parallel + N_\perp$:*

- *If $\|N_\perp\| \gg \|N_\parallel\|$: Error is detected with high probability (active protection dominates)*
- *If $\|N_\parallel\| \gg \|N_\perp\|$: State remains in E_λ, distortion bounded by stability (passive protection dominates)*
- *For intermediate noise: Both mechanisms contribute*

The key observation: **no error escapes both mechanisms**. Large errors are detected; small errors cause minimal distortion on stable eigenspaces.

4 Eigenspace Stability and State Fidelity

We now quantify how eigenspace stability translates to state protection.

4.1 Fidelity Under Perturbation

Definition 4.1 (State Fidelity)**.** For states $|\psi\rangle$ and $|\phi\rangle$, the fidelity is:

$$F(\psi, \phi) = |\langle\psi|\phi\rangle|^2 \tag{6}$$

Theorem 4.2 (Stability-Fidelity Bound)**.** *Let $|\psi\rangle \in E_\lambda$ be a code state and N be a perturbation with $\|N\|_2 \leq \epsilon$. Let δ be the spectral gap to adjacent eigenvalues. Then the state after projecting back to E_λ satisfies:*

$$F(\psi, \psi') \geq 1 - O\left(\frac{\epsilon^2}{\delta^2}\right) \tag{7}$$

where $|\psi'\rangle = P_\lambda N|\psi\rangle / \|P_\lambda N|\psi\rangle\|$.

Proof. By Davis-Kahan (Theorem 2.4), the eigenspace E_λ is perturbed by angle at most $\arcsin(\epsilon/\delta) \approx \epsilon/\delta$ for small ϵ.

The state $|\psi\rangle$ lies in E_λ. Under perturbation, the projection $P_\lambda N|\psi\rangle$ lies in the perturbed eigenspace E'_λ, which differs from E_λ by angle $O(\epsilon/\delta)$.

For a state in a subspace perturbed by angle θ:

$$F \geq \cos^2\theta \geq 1 - \theta^2 = 1 - O\left(\frac{\epsilon^2}{\delta^2}\right) \tag{8}$$

□

Corollary 4.3 (Stability Determines Protection)**.** *The fidelity loss scales inversely with the square of the spectral gap:*

$$1 - F \sim \frac{\epsilon^2}{\delta^2} \tag{9}$$

Graphs with larger spectral gaps provide stronger passive protection.

4.2 Beyond Spectral Gap: Eigenspace Stability

Corollary 4.3 suggests maximizing the spectral gap δ. The complete graph K_n has $\delta = n$, the maximum possible. But experiments show K_n has *poor* self-alignment (0.33 vs 0.90 for random regular graphs).

The resolution: **spectral gap bounds the worst case, but eigenspace structure determines the typical case**.

Proposition 4.4 (Structure Beyond Gap)**.** *Two phenomena reduce effective stability below the Davis-Kahan bound:*

1. ***Eigenspace degeneracy:*** *For degenerate eigenvalue λ (multiplicity > 1), individual eigenvectors rotate freely within E_λ even when E_λ itself is stable.*
2. ***Perturbation propagation:*** *In highly connected graphs, a local perturbation affects many edges, amplifying the effective ϵ.*

Example 4.5 (Complete Graph Failure)**.** In K_n:

- Eigenvalue $\lambda = n$ has multiplicity $n - 1$
- Any perturbation causes eigenvector rotation within the $(n-1)$-dimensional eigenspace
- A single-vertex perturbation changes $n - 1$ edges (every edge from that vertex)

The eigenspace is stable (Davis-Kahan), but individual encoded states are not.

Example 4.6 (Random Regular Graph Success)**.** In random d-regular graphs:

- Eigenvalues are generically simple (multiplicity 1)
- Eigenvectors are determined up to sign, not subject to rotation
- A single-vertex perturbation changes only d edges

Eigenvector stability directly measures structural stability.

5 Optimal Topology Selection

We now analyze which graph topologies provide optimal passive protection.

5.1 Comparison Framework

For a fair comparison, we evaluate:

1. **Spectral gap:** $\delta = \lambda_1$ (or gap around code eigenvalue)
2. **Eigenspace dimension:** $\dim(E_\lambda)$ determines encoding capacity
3. **Eigenvector stability:** $\mathbb{E}[|\langle v_i, v_i' \rangle|]$ under local perturbation
4. **Perturbation locality:** Number of edges affected by single-vertex perturbation

5.2 Experimental Results

Table 1: Topology Comparison for Passive Protection

Topology	n	Gap δ	Stability	Perturbation Size	Score
Random 4-regular	20	0.79	0.90	4	**0.63**
Complete K_{15}	15	15.0	0.33	14	0.59
Cycle C_{20}	20	0.10	0.65	2	0.47
Petersen	10	2.00	0.54	3	0.46
Hypercube Q_4	16	2.00	0.55	4	0.45
Star S_{15}	15	1.00	0.42	14	0.35

Theorem 5.1 (Optimal Topology)**.** *Random regular graphs achieve optimal balance of:*

1. ***Simple spectrum:*** *Eigenvalues are generically non-degenerate, so eigenvector stability measures true structural stability.*
2. ***Uniform structure:*** *Every vertex has equal degree, providing uniform "support" to eigenvectors.*
3. ***Local perturbations:*** *A single-vertex change affects only d edges (constant), not scaling with n.*
4. ***Averaging:*** *Random structure means no single edge is critical; perturbation effects average out.*

5.3 Scaling Analysis

Proposition 5.2 (Scaling of Protection)**.** *For d-regular graphs on n vertices:*

- ***Spectral gap:*** $\lambda_1 \geq d - 2\sqrt{d-1} + o(1)$ *for random d-regular graphs (Friedman's theorem [5]), approaching the Ramanujan bound.*
- ***Perturbation size:*** *Constant (d edges per vertex), independent of n.*
- ***Code dimension:*** $\dim(E_\lambda) \sim$ *constant for most eigenvalues (simple spectrum).*

This means: as n grows, random regular graphs maintain strong passive protection, while the code size (number of physical levels) grows without degrading stability.

6 Coherent Spectral Codes

We now define codes that satisfy both active and passive protection criteria.

6.1 Definition

Definition 6.1 (Coherent Spectral Code)**.** A *coherent spectral code* $\mathcal{C}(G, \lambda)$ is a spectral code satisfying:

1. **Detection criterion:** All single-vertex errors are detected (eigenvalue changes)
2. **Stability criterion:** Eigenspace stability $\mathbb{E}[|\langle v, v'\rangle|] \geq 1 - \eta$ for some threshold η
3. **Gap criterion:** Spectral gap $\delta \geq \delta_{\min}$ for some minimum gap

Theorem 6.2 (Coherent Code Protection)**.** *A coherent spectral code with stability $1 - \eta$ and gap δ provides:*

1. ***Active protection:*** *Errors with $\|N_\perp\| > \epsilon_{detect}$ are detected with probability $\geq 1 - e^{-c\epsilon^2_{detect}}$*
2. ***Passive protection:*** *For undetected errors, fidelity $F \geq 1 - \eta - O(\epsilon^2/\delta^2)$*

Proof. (1) follows from Part I: eigenvalue measurement detects the $N_\perp$ component.

(2) follows from Theorem 4.2: undetected errors have $N = N_\|$, which preserves eigenspace but may distort within it. Stability bounds this distortion by η, and Davis-Kahan provides the ϵ^2/δ^2 correction. □

6.2 Concrete Constructions

Example 6.3 (Random 4-Regular Code)**.** For random 4-regular graph G on $n = 20$ vertices:

- Code: $\mathcal{C}(G, \lambda_1)$ where $\lambda_1 \approx 0.79$
- Dimension: 1 (simple eigenvalue) — encodes 0 logical qubits but demonstrates principle
- Stability: 0.90
- Active protection: All single-vertex errors detected
- Passive protection: Undetected perturbations cause $\leq 10\%$ fidelity loss

Example 6.4 (Optimized Construction)**.** For practical encoding, use random d-regular graph with:

- Large n for code size
- Moderate d (e.g., $d = 4$ or $d = 6$) for balance of stability and gap
- Select eigenvalue λ with multiplicity ≥ 2 for logical encoding

Finding eigenvalues with multiplicity > 1 in random regular graphs requires careful construction (e.g., Cayley graphs of specific groups).

6.3 Comparison with Existing Approaches

Table 2: Protection Mechanisms Comparison

Approach	Active	Passive	Mechanism
Stabilizer codes	Yes	No	Syndrome measurement
Topological codes	Yes	Yes	Non-local encoding
Decoherence-free subspaces	No	Yes	Symmetry protection
Coherent spectral codes	**Yes**	**Yes**	**Eigenspace structure**

Coherent spectral codes provide both active and passive protection through eigenspace structure, complementing existing approaches.

7 Decoherence Dynamics

We now analyze how coherent spectral codes perform under continuous decoherence.

7.1 Decoherence Model

Consider Lindblad dynamics with local noise:

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_i \gamma_i \left(L_i \rho L_i^\dagger - \frac{1}{2}\{L_i^\dagger L_i, \rho\} \right) \tag{10}$$

where L_i are local (single-vertex) jump operators.

7.2 Coherence Time

Definition 7.1 (Coherence Time)**.** The coherence time T_2 is the timescale over which off-diagonal elements of ρ (in the code basis) decay to $1/e$ of their initial value.

Proposition 7.2 (Stability Enhances Coherence Time)**.** *For coherent spectral code with stability $s = 1 - \eta$:*

$$T_2 \sim \frac{1}{\gamma(1-s)} = \frac{1}{\gamma\eta} \tag{11}$$

where γ is the noise rate.

Sketch. Eigenspace stability means local noise operators have reduced effective strength when projected onto the code space. The projection reduces the noise rate by factor $(1 - s)$. □

Corollary 7.3 (Random Regular Advantage)**.** *Random regular graphs (stability 0.90) have predicted coherence time:*

$$T_2^{(RRG)} \sim \frac{1}{0.10\gamma} \tag{12}$$

Complete graphs (stability 0.33) have:

$$T_2^{(K_n)} \sim \frac{1}{0.67\gamma} \tag{13}$$

Ratio: random regular graphs achieve $\approx 6.7\times$ longer coherence time.

7.3 Numerical Prediction

Conjecture 7.4 (Coherence Enhancement)**.** For coherent spectral codes on random d-regular graphs with $d = O(1)$ and $n \to \infty$:

$$T_2 \sim \frac{\delta}{\gamma} \cdot f(d) \tag{14}$$

where $f(d)$ is a function of regularity satisfying $f(d) \to$ constant as d is fixed and $n \to \infty$.

This predicts that coherence time grows with spectral gap (which approaches the Ramanujan bound) while remaining stable in n.

8 Connection to Self-Reference

We briefly note the deeper connection to consciousness theory from [2].

8.1 Fixed Points and Stability

The Self-Aligning Topologies paper established:

- Eigenvectors are *fixed points* of self-modeling operators $M = g(L)$
- Stable self-representation requires topologies where these fixed points resist perturbation
- Random regular graphs provide optimal stability

8.2 Coherence as Self-Consistency

From this perspective, **quantum coherence is self-consistency of the encoded state**:

- A coherent state maintains its identity under perturbation
- Self-aligning topologies provide "support" for this identity
- Decoherence is failure of self-consistency

This suggests: the mathematical structure enabling stable self-reference (for consciousness) is the same structure enabling quantum coherence (for computation). Both require **fixed points that resist perturbation**.

8.3 Why Structured Randomness?

The finding that random regular graphs outperform symmetric structures has a conceptual interpretation:

- **Pure order** (complete graph): Every part connects to every other. A perturbation propagates everywhere. No localization.
- **Pure randomness** (Erdős-Rényi): No structure to maintain. Eigenvectors are unstable.
- **Structured randomness** (random regular): Local structure (fixed degree) with global diversity. Perturbations are absorbed locally.

This mirrors biological neural networks, which exhibit small-world topology: local clustering with random long-range connections.

9 Discussion

9.1 Summary of Results

1. **Dual protection:** Spectral codes provide both active (detection) and passive (stability) protection against errors.

2. **Stability-fidelity theorem:** Eigenspace stability translates directly to state fidelity bounds under perturbation.

3. **Optimal topology:** Random regular graphs achieve highest eigenspace stability, providing optimal passive protection.

4. **Coherent codes:** Codes on self-aligning topologies achieve predicted $\sim 6.7\times$ coherence time enhancement over complete graphs.

9.2 Limitations

1. **Encoding capacity:** Random regular graphs have generically simple spectra, limiting encoding to single states per eigenvalue. Practical codes require eigenvalues with multiplicity ≥ 2.
2. **Physical implementation:** Realizing the graph Laplacian as a physical Hamiltonian requires specific coupling patterns.
3. **Error correction:** We establish detection and passive resistance, not full correction. Recovery operations require further development.

9.3 Open Questions

1. Can we construct random-like graphs with controlled eigenvalue multiplicities for larger encoding?
2. How do coherent spectral codes compare to surface codes for realistic noise models?
3. Is there a fundamental limit relating code capacity, distance, and coherence time?
4. Can the self-aligning topology framework be extended to continuous-variable systems?

10 Conclusion

We have extended spectral codes by incorporating insights from self-aligning topology theory. The key results are:

1. **Dual protection framework:** Active error detection (eigenvalue measurement) complements passive protection (eigenspace stability).
2. **Stability-fidelity connection:** The Davis-Kahan theorem translates graph-theoretic stability into quantum information fidelity bounds.
3. **Topology optimization:** Random regular graphs provide optimal passive protection due to simple spectra, uniform structure, and local perturbation absorption.
4. **Coherence enhancement:** Coherent spectral codes on self-aligning topologies achieve predicted $\sim 6.7\times$ coherence time improvement.

This work bridges three domains:

- **Spectral graph theory:** Laplacian eigenstructure determines both detection and stability
- **Quantum error correction:** Eigenspaces provide natural code spaces with built-in protection
- **Self-reference theory:** The same mathematical structures enabling stable self-modeling enable quantum coherence

The observation that *structured randomness outperforms pure symmetry* for both self-reference stability and quantum coherence suggests a deep mathematical principle worthy of further investigation.

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